

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – COMPARATIVE ANALYSIS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4 – Precision (BL4)	Assessment:	Comparative Analysis
Weightage:	30%	Total Marks:	100
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
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Section / Batch:	Slot-B-2025	Date:	25.08.2026

Instructions to Students

- Identify the relevant concepts of L1, L2, and L3 cache, SRAM, DRAM, NAND Flash, MRAM, RRAM, virtual memory, paging, latency, bandwidth, and throughput.
- Analyze the memory requirements of different computer systems by considering cache levels, memory technologies, access latency, bandwidth, throughput, capacity, and energy consumption.
- Evaluate the performance characteristics of different memory technologies and determine their suitability for cache memory, main memory, secondary storage, and future hierarchical memory systems.
- Compute relevant performance measures such as access time, latency, bandwidth, throughput, hit/miss rate, average memory access time (AMAT), speedup, and energy efficiency wherever applicable.
- Interpret the results by assessing memory bottlenecks, performance improvements, technology trade-offs, and the suitability of each memory technology or memory hierarchy for a given application.

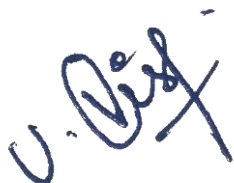
QUESTIONS

1. Evaluate L1, L2, and L3 cache levels based on access latency, bandwidth, and throughput. Analyze how each cache level influences processor performance and memory access efficiency.
2. Assess the performance characteristics of SRAM and DRAM with respect to access latency, bandwidth, and throughput. Justify the use of SRAM in cache memory and DRAM in main memory based on their respective characteristics.
3. Examine virtual memory, paging, and cache memory mechanisms in terms of access latency, memory utilization, and system throughput. Analyze how these mechanisms contribute to efficient memory management and overall system performance.
4. Analyze the performance characteristics of NAND Flash and DRAM with respect to access latency, bandwidth, and throughput. Evaluate their suitability for different levels of a hierarchical memory system and justify their applications.

5. Evaluate MRAM and RRAM technologies based on access time, throughput, energy efficiency, and endurance. Analyze their potential applications in future hierarchical memory systems and assess their performance advantages and limitations.

Code of Conduct

I Vishwa.U(192511192) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage(%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Marks Evaluation:

Q. No	Scope of Items (20)	Comparison Depth (25)	Use of Evidence (20)	Critical Thinking (20)	Clarity of Presentation (15)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 25	/ 20	/ 20	/ 15	/ 100

Course Coordinator Faculty Incharge

Question 1

Evaluate L1, L2, and L3 cache levels based on access latency, bandwidth, and throughput. Analyze how each cache level influences processor performance and memory access efficiency.

1. Overview

Modern processors use a multi-level cache hierarchy (L1, L2, L3) to bridge the widening speed gap between the CPU core and main memory (DRAM). Each level trades capacity for speed: the level closest to the core is the smallest and fastest, while levels farther away are larger, slower, and shared among more consumers.

2. Comparative Data

Attribute	L1 Cache	L2 Cache	L3 Cache (LLC)
Typical Capacity	32–64 KB (split I/D) per core	256 KB – 2 MB per core	8 – 32+ MB, shared across cores
Access Latency	~1–4 cycles (≈0.5–1 ns)	~10–14 cycles (≈3–7 ns)	~30–50 cycles (≈10–20 ns)
Bandwidth	Very high (100+ GB/s per core)	High, but shared read/write ports	Moderate; shared and subject to contention
Throughput	Highest — services almost every load/store	Reduces L1-miss penalty significantly	Filters most main-memory traffic before it reaches DRAM
Location / Scope	On-core, private, split instruction/data	On-core or per-core, unified	On-die, shared across all cores

3. Analysis — Influence on Processor Performance

- **Average Memory Access Time (AMAT):** The layered design directly reduces AMAT, computed recursively as: $AMAT = HitTime(L1) + MissRate(L1) \times [HitTime(L2) + MissRate(L2) \times (HitTime(L3) + MissRate(L3) \times MemoryAccessTime)]$. Because each level captures the majority of accesses missed by the level above (typically L1 hit rates of 90–97%, cascading misses filtered further by L2/L3), the effective latency the CPU experiences stays close to the L1 latency for most instructions.
- **Locality exploitation:** L1 is optimized for temporal and spatial locality of the currently executing instruction stream and working set; L2 captures a larger working set that no longer fits in L1; L3 captures cross-core sharing and streaming/irregular access patterns, reducing costly off-chip DRAM trips.
- **Bandwidth vs. contention trade-off:** L1's very high per-core bandwidth sustains multiple loads/stores per cycle, which is essential for superscalar/out-of-order execution. L3, despite lower latency-normalized bandwidth, must arbitrate among many cores, so its effective throughput per core is lower — this is why keeping data resident in L1/L2 (via good locality and prefetching) has an outsized performance impact.
- **Miss penalty amplification:** A miss at each level does not just add its own latency — it exposes the latency of the next level. A L1 miss serviced by L2 costs far less than one that ripples all the way to DRAM (often 100–300+ cycles), so L3's main performance contribution is preventing expensive off-chip trips rather than being fast in isolation.

4. Interpretation / Conclusion

Overall, L1 dominates instruction-level latency-sensitive performance, L2 acts as a buffer that captures the immediate working-set overflow from L1, and L3 acts as a last line of defense that maximizes system throughput by minimizing costly main-memory accesses and supporting inter-core data sharing. The three levels together form a latency-bandwidth-capacity trade-off curve that is fundamental to achieving low AMAT without the prohibitive cost of building all memory as fast, low-capacity SRAM.

Question 2

Assess the performance characteristics of SRAM and DRAM with respect to access latency, bandwidth, and throughput. Justify the use of SRAM in cache memory and DRAM in main memory based on their respective characteristics.

1. Overview

SRAM and DRAM are the two dominant volatile memory technologies used in the memory hierarchy. Their difference in cell design (6-transistor flip-flop vs. 1-transistor/1-capacitor) drives fundamentally different speed, density, cost, and power profiles.

2. Comparative Data

Attribute	SRAM	DRAM
Cell structure	6T (bistable flip-flop)	1T1C (transistor + capacitor)
Access Latency	~0.5–2 ns	~40–100 ns (incl. row activation, refresh overhead)
Bandwidth	Very high, sustained, multi-port capable	Lower; shared bus, burst-oriented (DDR)
Throughput	High — no refresh stalls, low miss-related delay	Lower — periodic refresh (~every 64 ms) and row-buffer conflicts interrupt access
Density / Cost per bit	Low density, high cost/bit	High density, low cost/bit
Power	No refresh power; leakage current in idle cells	Continuous dynamic refresh power
Typical Role	Cache memory (L1/L2/L3)	Main memory (DIMMs)

3. Justification

- **Why SRAM for cache:** Cache performance is judged almost entirely by hit latency; SRAM's sub-nanosecond access and absence of refresh cycles keep hit time predictable and minimal, which is essential since the CPU stalls on every cache access. Its low density is an acceptable trade-off because caches are intentionally small.
- **Why DRAM for main memory:** Main memory must hold gigabytes of code and data at reasonable cost; DRAM's high density and low cost per bit make large-capacity main memory economically feasible. Its higher latency is tolerable because the cache hierarchy filters the majority of CPU accesses before they reach DRAM, so DRAM latency affects only the smaller fraction of cache-miss traffic.
- **Systemic complementarity:** SRAM and DRAM are not competing but complementary: SRAM handles the latency-critical, small working set; DRAM handles the capacity-critical bulk storage. This division is what allows the overall system to approach SRAM-like effective latency at DRAM-like effective cost per byte.

4. Interpretation / Conclusion

The SRAM/DRAM split embodies the core principle of memory hierarchy design: exploit a small amount of fast, expensive memory to intercept most accesses (locality), backed by a large amount of slow, cheap memory to hold the full working set. Choosing SRAM for cache and DRAM for main memory is therefore not arbitrary but a direct consequence of matching technology characteristics to their positional role in the hierarchy

Question 3

Examine virtual memory, paging, and cache memory mechanisms in terms of access latency, memory utilization, and system throughput. Analyze how these mechanisms contribute to efficient memory management and overall system performance.

1. Overview

Cache memory and virtual memory (implemented via paging) are both hierarchical management mechanisms, but they operate at very different granularities and are managed by different agents — cache is hardware-managed at the cache-line level, whereas virtual memory/paging is managed cooperatively by the OS and the Memory Management Unit (MMU) at the page level.

2. Comparative Data

Aspect	Cache Memory	Virtual Memory / Paging
Granularity	Cache line (typically 64 bytes)	Page (typically 4 KB, up to 2 MB/1 GB huge pages)
Management	Hardware (LRU/pseudo-LRU replacement, hardware controller)	OS + MMU/TLB (page tables, page-replacement algorithms e.g. LRU, Clock)
Access Latency (hit)	~1–20 ns depending on level	TLB hit: ~1 cycle; page-table walk on TLB miss: tens of ns
Access Latency (miss)	Serviced by next cache level or DRAM (ns range)	Page fault → disk/SSD I/O, on the order of microseconds to milliseconds
Purpose	Reduce average latency to frequently used data	Provide a large, isolated, contiguous address space per process beyond physical RAM limits
Utilization impact	Exploits temporal/spatial locality of data	Exploits locality of reference at process level; enables memory over-commitment and multiprogramming

3. Analysis

- **Complementary hierarchy:** The Translation Lookaside Buffer (TLB) is itself a small, fast cache for page-table entries, meaning virtual memory relies on a cache-like mechanism to keep address translation fast; without it, every memory access would require a multi-level page-table walk.
- **Throughput contribution:** Cache memory improves throughput by keeping the CPU pipeline fed without frequent stalls on DRAM latency. Paging improves system-level throughput by allowing multiple processes to run concurrently with only their active working sets resident in physical RAM (demand paging), increasing multiprogramming and CPU utilization.
- **Cost asymmetry:** A cache miss costs nanoseconds; a page fault costs milliseconds (disk-class latency) — roughly five to six orders of magnitude higher. This asymmetry explains why OS page-replacement policies are far more sophisticated and conservative (e.g., working-set models, second-chance/Clock algorithms) than hardware cache-replacement policies, since a poor decision is vastly more expensive.
- **Memory utilization:** Paging allows non-contiguous physical allocation and over-subscription of memory (virtual address space larger than physical RAM), improving utilization efficiency across the whole system, while cache memory improves the utilization of already-resident data by keeping the CPU from idling on repeat memory accesses.

4. Interpretation / Conclusion

Cache memory and virtual memory/paging address different layers of the same underlying problem — the mismatch between CPU speed and memory latency/capacity. Cache optimizes fine-grained, hardware-speed reuse of data; paging optimizes coarse-grained, OS-level allocation and protection of memory. Together they let a system approach the illusion of memory that is simultaneously as fast as SRAM, as large as disk, and safely isolated per process.

Question 4

Analyze the performance characteristics of NAND Flash and DRAM with respect to access latency, bandwidth, and throughput. Evaluate their suitability for different levels of a hierarchical memory system and justify their applications.

1. Overview

NAND Flash and DRAM sit at different tiers of the memory-storage hierarchy: DRAM is volatile, byte-addressable working memory, while NAND Flash is non-volatile, block-addressable storage used mainly in SSDs. Their performance and cost characteristics are almost inverse, which is precisely why they serve different roles.

2. Comparative Data

Attribute	DRAM	NAND Flash
Volatility	Volatile (loses data without power)	Non-volatile (retains data without power)
Addressing	Byte-addressable	Block/page-addressable (program in pages, erase in blocks)
Read Latency	~40–100 ns	~25–100 μ s (roughly 1000 $\times$ slower than DRAM)
Write Latency	~40–100 ns	Hundreds of μ s; erase-before-write required, adding overhead
Bandwidth	High (tens of GB/s per channel)	Lower per-die, though SSDs aggregate many dies/channels
Endurance	Effectively unlimited (no wear-out)	Limited program/erase cycles (~1,000–100,000 depending on cell type: TLC/MLC/SLC)
Cost per bit	Higher	Lower, enabling large-capacity storage
Typical Role	Main memory (working set)	Secondary storage (SSD, persistent data)

3. Suitability Across the Hierarchy

- **DRAM as main memory:** Its nanosecond latency and byte addressability match the CPU's need for fast, fine-grained, random read/write access to active program data — it sits directly below the cache hierarchy, servicing cache misses.

- **NAND Flash as secondary storage:** Its non-volatility and low cost per bit make it ideal for persistent bulk storage where microsecond-scale latency is acceptable — e.g., holding the operating system, files, and data that must survive power loss, brought into DRAM/cache on demand via paging and I/O.
- **Emerging intermediate tiers:** The multi-order-of-magnitude latency gap between DRAM and NAND Flash has motivated storage-class memory technologies (e.g., 3D XPoint-class NVMe devices) positioned between the two, offering near-DRAM speed with NAND-like persistence — illustrating how hierarchical design continues to evolve to close latency/cost gaps.
- **Write-endurance trade-off:** NAND's limited program/erase endurance rules it out for roles requiring constant overwriting (like a cache or main memory), reinforcing why it is confined to storage tiers where wear-leveling algorithms can amortize writes across the device.

4. Interpretation / Conclusion

DRAM and NAND Flash are optimized for opposite ends of the speed-persistence-cost spectrum. DRAM's advantage — speed — suits it to the volatile, latency-critical main-memory tier, while NAND Flash's advantages — non-volatility and density/cost — suit it to the persistent, capacity-critical storage tier. A well-designed hierarchical system relies on the pairing of DRAM and NAND (often mediated by an intermediate cache/DRAM buffer inside the SSD itself) to balance speed, persistence, and cost.

Question 5

Evaluate MRAM and RRAM technologies based on access time, throughput, energy efficiency, and endurance. Analyze their potential applications in future hierarchical memory systems and assess their performance advantages and limitations.

1. Overview

MRAM (Magnetoresistive RAM) and RRAM (Resistive RAM) are emerging non-volatile memory (NVM) technologies often discussed as candidates for 'universal memory' — combining DRAM-like speed with Flash-like persistence. They differ in their underlying physical switching mechanism, which shapes their respective strengths

2. Comparative Data

Attribute	MRAM	RRAM (ReRAM)
Switching Mechanism	Spin-transfer torque (STT) alters magnetic orientation of a magnetic tunnel junction (MTJ)	Formation/rupture of a conductive filament changes resistance of a metal-oxide layer
Access Time	~5–20 ns (near-DRAM/SRAM speed)	~10–50 ns (comparable, slightly more variable)
Throughput	High; supports frequent read/write with low latency variance	High for reads; writes can be less deterministic due to filament formation variability
Energy Efficiency	Low write energy, no refresh needed (non-volatile)	Very low write energy; simple two-terminal structure aids low-power design

Attribute	MRAM	RRAM (ReRAM)
Endurance	Very high ($\sim 10^{12}$ – 10^{15} cycles), near-DRAM-class	Moderate ($\sim 10^6$ – 10^{12} cycles depending on material), lower than MRAM
Density / Scalability	Moderate; cell size limited by MTJ fabrication	Potentially very high density; simple structure suits 3D stacking and crossbar arrays
Non-volatility	Yes	Yes

3. Analysis of Future Applications

- **MRAM in the hierarchy:** Its combination of near-SRAM speed, very high endurance, and non-volatility makes it a strong candidate for replacing SRAM in L2/L3 cache or as non-volatile working memory — enabling instant-on systems and reducing static leakage power in idle cache lines, while withstanding the very high write frequency caches experience.
- **RRAM in the hierarchy:** RRAM's higher potential density and crossbar-compatible structure make it attractive for storage-class memory bridging DRAM and Flash, and for emerging in-memory/neuromorphic computing, where its analog resistance states can represent synaptic weights directly, enabling compute-in-memory architectures that reduce data movement.
- **Advantages:** Both eliminate the volatility/refresh-power overhead of DRAM and the slow, high-latency writes of NAND Flash, positioning them as candidates to collapse multiple hierarchy levels (e.g., a unified fast, persistent memory tier) and enable rapid checkpoint/restore and energy-proportional computing.
- **Limitations:** MRAM's scaling is constrained by MTJ fabrication complexity and moderate density compared to DRAM/Flash, limiting near-term use to smaller, high-value tiers such as cache or embedded NVM. RRAM's write variability and comparatively lower endurance currently restrict it from replacing SRAM/DRAM in latency- and endurance-critical roles, positioning it more realistically as a storage-class or accelerator memory rather than a cache replacement today.

4. Interpretation / Conclusion

MRAM and RRAM both push toward the long-term goal of a 'universal memory' that merges the speed of SRAM/DRAM with the persistence of Flash, but they are presently complementary rather than interchangeable: MRAM's endurance and speed suit endurance-critical, latency-sensitive tiers (cache/near-memory), while RRAM's density and analog switching suit capacity-oriented storage-class memory and in-memory computing. Their adoption in future hierarchical systems will likely be tier-specific rather than a wholesale replacement of existing SRAM/DRAM/Flash layers.